

MULTI-LAYER FEATURE AND PART-TOKEN FUSION FOR URBAN OBJECT RE-IDENTIFICATION

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ABSTRACT

Urban object re-identification is challenging because mass-produced objects are near-identical, yet a single object’s appearance shifts with season, viewpoint, and lighting. On the URBAM-ReID2026 dataset of traffic signs, crosswalks, containers, and rubbish bins, we adapt a Part-Aware Transformer with a Vision Transformer backbone. Our system combines stochastic augmentation with AdamW optimisation, multi-layer classification token and part-token fusion, traffic sign classification and class-aware re-ranking. Across over 50 configurations, we improve mean Average Precision from 0.101 to 0.164 (+62.6% relative) and report negative results that inform future work.

Index Terms— Re-Identification, Vision Transformers, Urban Infrastructure, Smart Cities.

1. INTRODUCTION

Re-identification (ReID) of static urban infrastructure is relevant to smart-city asset tracking, yet under-explored compared to person [1] ReID. Key challenges are low inter-class variability and strong appearance changes across viewpoints, lighting, and seasons. The URBAM-ReID2026 dataset [2] pairs reverse-direction queries (camera c004) with forward-direction galleries (c001–c003), producing $\sim 180^\circ$ viewpoint flips.

We build on the Part-Aware Transformer (PAT) [3], which prepends three learnable part tokens to the Vision Transformer (ViT) [4] patch sequence; each attends to a masked spatial region. Our contributions: (1) multi-layer classification (CLS) token fusion with part-token concatenation; (2) stochastic augmentation [5] with AdamW [6] and higher-resolution training; (3) class-aware k -reciprocal re-ranking; and (4) negative-result analysis covering ensembles, Camera-Aware Jaccard (CAJ), Traffic Sign Classifier and prototype filtering.

2. METHODOLOGY

Baseline. PAT with ViT-Large (24 layers, 1024-dim, 307M params) pre-trained on ImageNet-21k. Three part tokens attend to overlapping horizontal bands via binary masks

$\mathbf{M} \in \{0, 1\}^{N \times N}$. Training uses identity cross-entropy (label smoothing $\epsilon = 0.1$), triplet loss, and patch-clustering loss [3]. UrbAM-ReID [7] is merged for additional training diversity.

Stochastic augmentation and optimiser. Following [5], each image randomly receives 0, 1, or 2 augmentations ($p = 0.15, 0.45, 0.40$) from a pool of perspective distortion, brightness/contrast, rotation, and zoom. Colour Jitter and Random Patch Transfer ($p = 0.3$) are applied on top; Local Grayscale Transformation (LGT) and Random Erasing are disabled. Disabling LGT was critical because traffic-sign subtype information is strongly colour-dependent. We use AdamW [6] at 5×10^{-4} and increase resolution to 320×192 .

CLS fusion with part-token concatenation. Earlier ViT layers retain texture; deeper layers encode semantics. At inference we concatenate CLS tokens from the last K layers, then append the three part tokens from the final layer and ℓ_2 -normalise:

$$\mathbf{f} = \frac{[\text{CLS}_{L-K+1}; \dots; \text{CLS}_L; \text{Part}_1^L; \text{Part}_2^L; \text{Part}_3^L]}{\|\cdot\|_2} \quad (1)$$

With $K = 8$ this yields $(8 + 3) \times 1024 = 11264$ dimensions at *zero retraining cost*. The part tokens encode local region-specific features that the global CLS token does not capture, providing complementary information for retrieval.

Class-aware post-processing. A hard penalty ($\alpha = 10^6$) eliminates cross-class matches. Per-class k -reciprocal re-ranking [8] parameters (k_1, k_2, λ) are tuned via staged grid search on UrbAM-ReID, followed by CAJ scaling. For traffic signs, we additionally train a lightweight semantic sign classifier with 15 visually grouped categories (e.g. warning, prohibitory, directional, and informational signs). During retrieval, cosine distances are softly penalised according to classifier probability disagreement, reducing semantically implausible matches while preserving ranking flexibility.

3. EXPERIMENTS

Dataset. URBAM-ReID2026 [2]: 4,961 images, 397 identities, four categories. Combined with UrbAM-ReID [7], training totals 17,562 images / 1,453 identities. Results: mAP@100 on the Kaggle public leaderboard (49% of test).

Ablation (Table 1). Stochastic augmentation + AdamW + higher resolution gives the largest gain (+1.3 pts), confirm-

Table 1. Cumulative ablation on the Kaggle public leaderboard.

Configuration	mAP	Δ
PAT ViT-L baseline (SGD, 256×128)	.101	—
+ External data (UrbAM-ReID)	.120	+.019
+ Hard class filtering	.123	+.004
+ CLS fusion ($K=8$)	.139	+.016
+ RPT + CJ (LGT off)	.148	+.008
+ Stochastic aug + AdamW + 320×192	.160	+.013
+ Part-token concatenation	.164	+.004
<i>Negative results (vs. best system)</i>		
Class-conditioned training	.086	−.078
Query expansion	.087	−.077
Camera feature normalisation	.118	−.046
Model ensemble (SGD + AdamW)	.144	−.020
Prototype filtering (GTSRB ViT)	.140	−.024
Traffic Classifier (TC)	.154	−.010

Table 2. Per-class performance (UrbAM-ReID validation).

Class	Queries	mAP	Rank-1
Traffic Signs	318 (61%)	.211	17.3%
Crosswalks	88 (17%)	.278	25.0%
Rubbish Bins	20 (4%)	.516	40.0%
Containers	91 (18%)	.574	31.9%

ing data-level strategies dominate in low-data regimes [5]. CLS fusion adds +1.6 mAP at zero retraining cost. Part-token concatenation recovers +0.4 mAP of local information PAT normally discards. Batch size 64 is optimal. Epoch 25 is best for the SGD model; 100 epochs degrades to .135 mAP.

Per-class (Table 2). Traffic signs (61% of queries) achieve the lowest mAP (.211) due to subtype ambiguity—prohibitory, warning, mandatory, and directional signs share one embedding space—and rear-view unrecognisability from the 180° viewpoint flip. Containers achieve the highest mAP (.574) owing to distinctive manufacturer markings and surface damage. Crosswalks (.278) are limited by the direction reversal, which presents stripes from the opposite end. Rubbish bins (.516) benefit from compact size and distinctive lid shapes despite frequent occlusion.

Negative results. Class-conditioned training confuses identity discrimination. Query expansion propagates noisy rankings. Camera normalisation removes discriminative cues alongside bias. Ensembling SGD and AdamW models was ineffective likely due to feature-space mismatch. Hard prototype filtering of traffic signs with a GTSRB classifier failed due to strong inter-prototype similarity in the embedding space. A lightweight traffic-sign classifier with 15 manually grouped semantic categories produced modest gains through soft probabilistic filtering, but became ineffective when combined with strong augmentation and AdamW.

4. CONCLUSION

We achieve 62.6% relative mAP improvement via stochastic augmentation, AdamW at higher resolution, CLS fusion with part-token concatenation, and class-aware re-ranking. Our analysis yields three key findings: (1) data-level strategies (augmentation, resolution) dominate model-level changes in low-data urban ReID; (2) PAT part tokens contain complementary local information that improves retrieval when concatenated at inference; (3) post-processing parameters and ensemble strategies that work individually do not necessarily compose, making per-configuration tuning essential. Traffic signs remain the primary bottleneck (.211 mAP, 61% of queries); future work will explore OCR-guided filtering and self-supervised features.

5. REFERENCES

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